Describe a wave

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Identify amplitude, wavelength, frequency and period.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 11.*
15°C; ΔT is proportional to absorbed energy under these conditions.

02. Worked example: explain the model
Amplitude vertical from equilibrium, wavelength horizontal crest spacing; period belongs on time axis.

03. Read and interpret the model
Crest-to-trough is 2 A for a symmetric sinusoid; wavelength matches successive crests.

04. Apply the physics
First trial: f = 6/5 = 1.2 Hz and T = 5/6 ≈ 0.833 s. Second trial: T = 0.5 s and f = 2 Hz, so it oscillates more often.

05. Explain your reasoning
No; maximum displacement and cycles per second are different independently variable properties in the basic model.

06. Review: connect an earlier idea *Review: Unit 5, Day 2.*
Q=mc ΔT=19200 J, assuming the supplied energy heats the sample.

07. Exit check
3 cycles;0.5 Hz.

08. STRETCH: test the model
Vertical excursion differs, crest spacing unchanged. Amplitude alone does not determine frequency.

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 11.*
Absorbed 7500 J; source 150000 J.

02. Worked example: explain the model
Amplitude vertical from equilibrium, wavelength horizontal crest spacing; period belongs on time axis.

03. Read and interpret the model
Crest-to-trough is 2 A for a symmetric sinusoid; wavelength matches successive crests.

04. Apply the physics
25 Hz;0.040 s;0.040 m amplitude.

05. Explain your reasoning
No; maximum displacement and cycles per second are different independently variable properties in the basic model.

06. Review: connect an earlier idea *Review: Unit 5, Day 2.*
$c=Q/(m\,\Delta T)=1000$ J/(kg·°C). No phase change and no unaccounted loss/input.

07. Exit check
$\lambda=0.50$ m; f 10 Hz.

08. STRETCH: test the model
$f=2.5$ Hz; $T=0.4$ s; $v=3$ m/s.